Week 2: Healthcare Vertical Assignment

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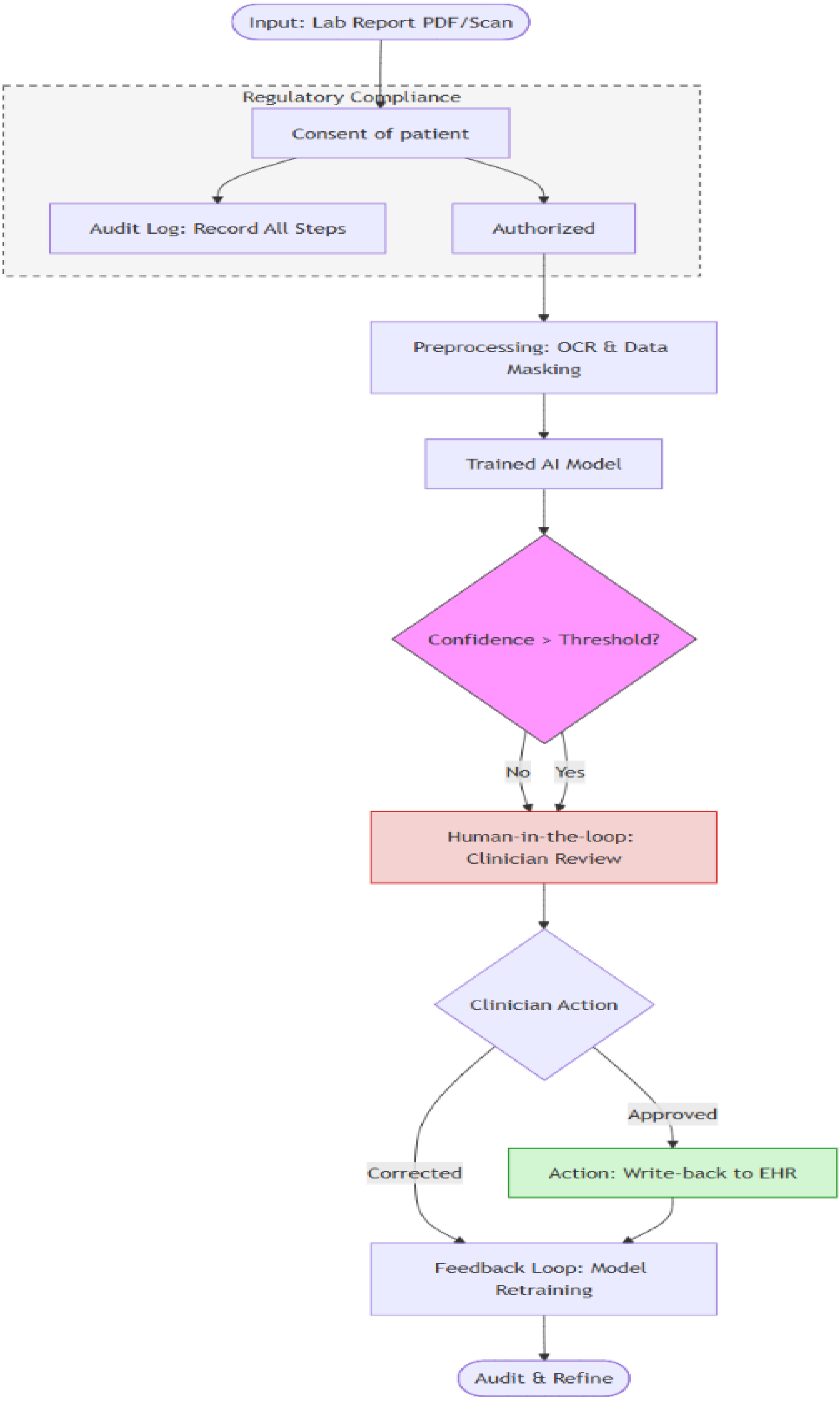
Part A: Pain-Point Map

Pain Point	Business Impact	AI-Solvable?	Use Case Type	Regulatory Constraints
Manual Data Entry	High admin cost, delays in patient care delivery.	Yes	NLP / OCR (Extraction)	HIPAA/DPDP: Consent & de-identification required.
Transcription Errors	Patient safety risk due to incorrect treatment.	Yes	NLP (Summarization)	Every automated decision must be recorded.
Delayed Decision Support	Missed clinical indicators; delayed intervention for at-risk patients.	Yes	Predictive Analytics / Pattern Recognition	Clinical explainability and mandatory human validation required.

Part B: Solution Flowchart


Input: Lab Report PDF/Scan
Regulatory Compliance
Consent of patient
Audit Log: Record All Steps
Authorized
Preprocessing: OCR & Data Masking
Trained AI Model
Confidence > Threshold?
No
Yes
Human-in-the-loop: Clinician Review
Clinician Action
Approved
Corrected
Action: Write-back to EHR
Feedback Loop: Model Retraining
Audit & Refine

Flow Logic:

1. **Input:** Raw lab report PDF/scan.
2. **Consent & De-identification Gate:** Data is masked/minimized before processing.
3. **Preprocessing:** OCR converts scans to text.
4. **Model:** NLP/LLM extracts and summarizes key findings.
5. **Decision Point:** Confidence threshold check; routes to clinician if low confidence.
6. **Human-in-the-loop:** Clinician reviews and signs off on the AI draft.
7. **Action:** Note is written back to the EHR.
8. **Audit/Feedback:** Every step is logged. Clinician corrections improve the model.

Part C: Justification Note

Why this technique? The challenge of converting unstructured lab reports into structured clinical data is best solved by combining OCR for text digitization and NLP/LLM for intelligent summarization. This matches the problem shape of document intelligence where variability in lab formats is high.

Data & Consent: Data lives in the diagnostic lab's storage and the hospital's EHR. Consent is obtained at the point of service and must be logged as part of the digital workflow. We ensure compliance by masking PII (de-identification) at the very start of the pipeline.

Top 3 Risks:

1. **Automation Bias:** Clinicians might over-rely on AI summaries.
   Mitigation: Mandatory clinician sign-off is non-negotiable.
2. **Privacy Breach:** Potential PII exposure during processing.
   Mitigation: Strict de-identification gates before the model sees data.
3. **Model Drift:** Performance degradation over time.
   Mitigation: Ongoing audit logs and feedback loops to monitor accuracy.

Minimum Viable Product (MVP): The MVP will focus on a single department like general medicine to ensure the clinician has full oversight while the system is refined.